

Problem Solving with AI (Claude Code + JavaScript)

Q1. JavaScript Coding Practice

1. Convert Celsius to Fahrenheit

- **Problem Statement:**

Write a function that takes a temperature in Celsius and converts it to Fahrenheit using the formula:

$$F = (C \times 9/5) + 32$$

.

- **Source Code:**

codeJavaScript

```
// Function to convert Celsius to Fahrenheit
function celsiusToFahrenheit(celsius) {
    let fahrenheit = (celsius * 9 / 5) + 32;
    return fahrenheit;
}

// Test cases / Execution
let tempC = 25;
let convertedF = celsiusToFahrenheit(tempC);
console.log(`${tempC}°C is equal to ${convertedF}°F`);
```

- **Input & Output:**

- **Input:** 25 (Celsius)
- **Output:** 77 (Fahrenheit)

```
[Running] node "e:\Module_5_Assignment.js"
25°C is equal to 77°F
```

```
[Done] exited with code=0 in 0.228 seconds
```

- **Explanation:**

The function takes a numeric parameter representing Celsius. It applies the mathematical conversion formula by multiplying the Celsius value by 9/5 (or 1.8) and adding 32, then returns the final computed Fahrenheit value using template literals.

2. Find the Factorial of a Number

- **Problem Statement:**

Write a function that calculates the factorial of a given non-negative integer (e.g.,

$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

).

- **Source Code:**

codeJavaScript

```
// Function to calculate factorial
function findFactorial(n) {
  if (n < 0) return "Factorial not defined for negative numbers";
  let factorial = 1;
  for (let i = 1; i <= n; i++) {
    factorial *= i;
  }
  return factorial;
}

// Test cases / Execution
let num = 5;
console.log(`The factorial of ${num} is:`, findFactorial(num));
```

- **Input & Output:**

- **Input:** 5
- **Output:** 120

```
[Running] node "e:\Module_5_Assignment.js"
The factorial of 5 is: 120
```

```
[Done] exited with code=0 in 0.275 seconds
```

- **Explanation:**

We initialize a variable `factorial` to 1. A `for` loop runs from 1 up to the given number `n`, successively multiplying the running product by each integer in the sequence. Edge cases like negative numbers are handled via an initial check.

3. Check for Palindrome

- **Problem Statement:**

Write a function that checks if a given word (e.g., "racecar") reads the same backward as forward and returns `true` or `false`.

- **Source Code:**

codeJavaScript

```
// Function to check if a word is a palindrome
function isPalindrome(str) {
  // Clean string: convert to lowercase to handle case sensitivity
  let cleanStr = str.toLowerCase();
  let reversedStr = cleanStr.split('').reverse().join('');
  return cleanStr === reversedStr;
}

// Test cases / Execution
console.log(isPalindrome("racecar")); // true
console.log(isPalindrome("Hello"));  // false
```

- **Input & Output:**

- **Input:** "racecar"
- **Output:** true

```
[Running] node "e:\Module_5_Assignment.js"
true
false

[Done] exited with code=0 in 0.323 seconds
```

- **Explanation:**

The function first normalizes the input string by converting it to lowercase. It then splits the string into an array of characters, reverses the array order, and joins it back into a string. Finally, it checks strict equality (===) between the original cleaned string and the reversed string, returning `true` if they match and `false` otherwise.

4. Sum of Array Elements

- **Problem Statement:**

Write a function that takes an array of numbers and returns the sum of all elements in the array.

- **Source Code:**

codeJavaScript

```
// Function to sum array elements
function sumOfArray(arr) {
  let sum = 0;
  for (let i = 0; i < arr.length; i++) {
    sum += arr[i];
  }
  return sum;
}

// Test cases / Execution
let numbers = [10, 20, 30, 40, 50];
console.log("Array:", numbers);
console.log("Sum of elements:", sumOfArray(numbers));
```

- **Input & Output:**

- **Input:** [10, 20, 30, 40, 50]
- **Output:** 150

```
[Running] node "e:\Module_5_Assignment.js"
```

```
Array: [ 10, 20, 30, 40, 50 ]
```

```
Sum of elements: 150
```

```
[Done] exited with code=0 in 0.192 seconds
```

- **Explanation:**

We define a accumulator variable `sum` starting at 0. A `for` loop iterates through each index of the array, adding the value of each element (`arr[i]`) to the accumulator. Once the loop finishes, the total sum is returned.

5. FizzBuzz (Basic)

- **Problem Statement:**

Write a loop from 1 to 15 that prints "Fizz" for numbers divisible by 3, "Buzz" for numbers divisible by 5, "FizzBuzz" for numbers divisible by both, and the number itself otherwise.

- **Source Code:**

codeJavaScript

```
// Function for FizzBuzz logic from 1 to 15
function runFizzBuzz() {
    for (let i = 1; i <= 15; i++) {
        if (i % 3 === 0 && i % 5 === 0) {
            console.log("FizzBuzz");
        } else if (i % 3 === 0) {
            console.log("Fizz");
        } else if (i % 5 === 0) {
            console.log("Buzz");
        } else {
            console.log(i);
        }
    }
}

// Execution
runFizzBuzz();
```

- **Input & Output:**

- **Input:** Numbers from 1 to 15 (loop range)
- **Output:**

codeText

```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
```

```
[Running] node "e:\Module_5_Assignment.js"
```

```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
```

- **Explanation:**
The loop checks conditions sequentially. **Crucially**, the combined condition (`i % 3 === 0 && i % 5 === 0`) for "FizzBuzz" is evaluated *first* so that numbers like 15 are caught before triggering just "Fizz" or "Buzz". Then it checks individual divisibility using the remainder operator (%).